



Nikolaos Rotas

SOFTWARE ENGINEER

Patras, Greece

📞 +30 6982994556 | ✉️ rotasn@gmail.com | 🏠 rotasn.github.io | 📷 [rotasn](#) | 🌐 [rotasn](#)

Summary

Motivated and resilient software engineer with a strong foundation in computer engineering and a passion for building real-world applications. Experienced in both solo and team projects, including desktop apps, automation, and game-related tooling. Eager to grow as a developer and contribute to meaningful software solutions.

Technical Skills

JavaScript, TypeScript, Angular, React, Electron, Python, Ionic, Algorithms, NLP

Projects

Marauder — Interactive Fiction Auto-Mapper github.com/rotasn/marauder

Auto-generates maps for text-based adventure games using rule-based NLP and traversal algorithms.

- Implemented auto-mapping using rule-based parsing
- Quick travel with BFS; auto-exploration with DFS
- Built layered UI; implemented save/load in JSON

Ball on Beam Simulator github.com/rotasn/ball-on-beam

Interactive simulation of ball-and-beam control using PID logic

- Built with React
- Implemented real-time PID loop to stabilize ball position
- Visual feedback with SVG-rendered beam and ball

File Encryptor github.com/rotasn/file-encryptor

Simple GUI-based file encryption tool.

- Built with PyQt5 and Python cryptography
- Fernet symmetric encryption and PBKDF2
- Password strength checker + backup support

Connect 4 GUI github.com/rotasn/connect4

Playable GUI-based Connect 4 game with basic AI

- Implemented AI opponent
- Mouse/keyboard controls + visual enhancements

Work Experience

Frontend Developer — University of Patras Networks and Architectures Group
July 2025 – now (Patras, Greece)

- Developing projects for the Smart Cities Team
- Responsible for web and mobile versions of SenseCity and SafeAmea

Senior Data Curator (Freelance) — Veeva Systems (via Upwork)
Dec 2021 – Mar 2023 (Remote)

- Maintained 98% data accuracy
- Collaborated with international teams

Intern — Informatics Department, University Hospital of Patras
Jun 2021 (Patras, Greece)

- Assisted IT support
- Gained exposure to hospital information systems and networks

Education

Integrated Master's (MSci) in Electrical and Computer Engineering

University of Patras — Graduated March 2025

- Focus: Software, Algorithms, Control Systems
- Thesis: Development of Interactive Fiction Mapping Application

Certifications

 freeCodeCamp – Front End Development Libraries

- React, Redux, jQuery, Bootstrap, SASS

 Cisco Networking Academy – NGN Linux Unhatched

- Linux fundamentals, permissions and command line basics

Languages

Greek - Native

English - Fluent (Cambridge FCE)

German - Basic (Goethe B1)

Soft Skills

Teamwork

Problem-solving

Adaptability

Interests

Gardening

Aquascaping

Animal rescue and stray animal support